p-adic Analysis 2

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April 2026

Problem 1

Repeatedly solving $7a_n \equiv 1 \pmod{3^n}$ gives us

$$\frac{1}{7} = \ldots 01212010212011_3.$$

Problem 2

Yes, since $5 \not\equiv 0 \pmod{11}$, there exists 5^{-1}. Letting $x = 5^{-1} \cdot 3$, we have that $5x = 3$.

Problem 3

We need $x^2 \equiv -1 \pmod{5}$, which is satisfied by any $x \equiv 2, 3 \pmod{5}$. Now, assume for some $n \geq 1$ we found x_n where

$$x_n^2 \equiv -1 \pmod{n}.$$

We want $x_{n+1} = x_n + a \cdot 5^n$ for some $a \in \{0, 1, 2, 3, 4\}$. So

$$x_{n+1}^2 = x_n^2 + 2x_n a 5^n + a^2 5^{2n}.$$

Taking this mod5^{n+1} gives us

$$x_n^2 + 2x_n a 5^n \equiv -1 \pmod{5^{n+1}}.$$

Since $x_n^2 \equiv -1 \pmod{5^n}$, we can write $x_n^2 + 1 = c \cdot 5^n$ for some $c \in \mathbb{Z}$. Then we get

$$c \cdot 5^n + 2x_n a 5^n \equiv 0 \pmod{5^{n+1}},$$
$$c + 2x_n a \equiv 0 \pmod{5},$$

which has a unique solution $a \pmod{5}$.

The first 10 digits are 2121342303.

There is no solution in $\mathbb{Z}_7$ because all the squares mod7 are

$$0^2 \equiv 0, 1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 2, 4^2 \equiv 2, 5^2 \equiv 4, 6^2 \equiv 1,$$

none of which are 6.

Problem 11

These are just the rational numbers that have nonnegative p-adic valuation, which are rationals of the form a/b where $p \nmid b$.

Problem 12

Let $x \in \mathbb{Q}_p^\times$. There is a unique $n \in \mathbb{Z}$ such that $x = p^n y$ with $y \in \mathbb{Z}_p^\times$. So the problem reduces to writing $y = ru$ with $r \in \{1, \ldots, p-1\}$ and $u \in 1 + p\mathbb{Z}_p$.

Since $y \in \mathbb{Z}_p^\times$, there is a unique $r \in \{1, \ldots, p-1\}$ with

$$y \equiv r \pmod{p}.$$

Then r is a p-adic unit as well, so $r^{-1} \in \mathbb{Z}_p$. Define

$$u = r^{-1} y,$$

so that $y = ru$ and $x = p^n ru$. Since $y \equiv r \pmod{p}$, we get

$$u \equiv 1 \pmod{p},$$

as desired. This is unique because both n and r were forced to be unique.

Problem 15

The set $\mathbb{Q}$ is already complete with respect to the trivial absolute value because Cauchy sequences are eventually constant rational numbers, so they always converge to rational numbers.

Problem 16

We have that

$$\prod_p |x|_p = \prod_p p^{-v_p(x)} = \left(\prod_p p^{v_p(x)} \right)^{-1}.$$

But by definition,

$$\prod_p p^{v_p(x)} = |x|$$

since it is the exact prime factorization of $|x|$. Since $|x| = |x|_\infty$,

$$|x|_\infty \prod_p |x|_p = |x|_\infty \cdot |x|^{-1} = 1.$$

Problem 17

Assume $x = r^2$ for some $r \in \mathbb{Q}$. Since $r \in \mathbb{R}$ and $r \in \mathbb{Q}_p$, $x = r^2$ is a perfect square in $\mathbb{R}$ and $\mathbb{Q}_p$.

Now, assume x is a square in $\mathbb{Q}_p$ for every p, and also a square in $\mathbb{R}$ (which forces it to be positive and finite). Fix a prime p. Since x is a square in $\mathbb{Q}_p$, there exists $y_p \in \mathbb{Q}_p$ such that $x = y_p^2$, and therefore $v_p(x) = 2v_p(y)$. So $v_p(x)$ is even for all p. Now, write

$$x = \prod_p p^{e_p}.$$

Since each e_p is even, we can let $f_p = e_p/2$ and write

$$x = \left(\prod_p p^{f_p} \right)^2,$$

so x is a perfect square in $\mathbb{Q}$.

Problem 18

Part (b)

Consider

$$\mathbb{R} = \bigcup_{n=1}^{\infty} (-n, n).$$

Any finite subcover is of the form

$$(-n_1, n_1) \cup \ldots \cup (-n_k, n_k) = (-N, N)$$

where $N = \max_i(n_i)$, and this doesn't cover $\mathbb{R}$.

Part (c)

First assume A is compact. It has to be unbounded because otherwise,

$$U_n = (-n, n), \quad n \geq 1$$

would form an open cover of A with no finite subcover. To show A is closed, let $x \in \mathbb{R} \setminus A$. For each $a \in A$, set

$$U_a = \left(a - \frac{1}{2}|x - a|,\ a + \frac{1}{2}|x - a| \right).$$

Then $\{X_a\}_{a \in A}$. By compactness, only finitely many are required:

$$A \subseteq U_{a_1} \cup \cdots \cup U_{a_n}.$$

Let

$$\varepsilon = \min_i \frac{|x - a_i|}{2} > 0.$$

Then $(x - \varepsilon, x + e)$ doesn't intersect with any of the U_{a_i}, so it misses A. So $R \setminus A$ is open, and therefore A is closed.